The Case Story

One case, told start to finish — Layers 1 through 5

Write this the way you'd explain the case to a colleague: start with the patient, end with how a wrong answer should be caught. Plain language throughout — no code, JSON, or YAML required. Engineering turns your words into the evaluation system.

1. **The Patient — what the model is shown (history, labs, imaging, pathology)**
2. **The Question — what we ask the model to figure out**
3. **The Reasoning — how an expert thinks it through, step by step**
4. **The Answer — the reference recommendation**
5. **The Grading — how a right vs. dangerous-wrong answer is told apart**

Case label (engineering can assign if unsure)
e.g. dlbcl_0006_cutaneous — short title

Difficulty
beginner/intermediate/advanced

Layer 1 — The Patient

This becomes exactly what the model is allowed to see. Nothing from later layers leaks in here.

How they presented
Age, sex, presentation, symptoms, performance status (ECOG). No real patient data.

What the labs showed
CBC, chemistry, LDH, and any other relevant labs with values and units.

What the imaging showed
Modality (e.g. PET-CT) and findings: nodal sites, size, extranodal/organ involvement.

What the pathology showed
Biopsy result, IHC markers with %, FISH rearrangement results, bone marrow status.

Synthetic training case for AI evaluation only. Not clinical decision support. Do not enter real patient-identifiable information.

Layer 2 — The Question

This is the standard task we put to the model for every case. Adjust only if this case needs something different.

The model is asked, using only the Layer 1 findings, to:

1. Establish the diagnosis and biological subtype.
2. Determine the stage.
3. Determine the molecular/biological classification (e.g. double-expressor vs. double-hit).
4. Calculate the appropriate prognostic score(s).
5. Recommend first-line treatment.
6. Assess CNS-relapse risk and whether prophylaxis should be considered.
7. Recommend a response-assessment strategy, and name any missing information.

Anything different for this case? (optional)

e.g. a competing diagnosis to rule out, or data deliberately withheld to raise difficulty.

Layer 3 — The Reasoning

Before giving the final answer — how would you think it through, out loud, step by step?

Bullet points are fine. This is the reasoning path, not the final recommendation (that's Layer 4).

Diagnosis & cell-of-origin

e.g. what the biopsy establishes; which IHC markers point to which Hans call, and its limits.

Staging

e.g. which sites put this above/below the diaphragm; how organ involvement changes the stage.

Molecular classification

e.g. IHC overexpression values vs. FISH result, and how they combine to a classification.

Prognosis

e.g. which score(s) apply and which supplied facts feed them.

First-line treatment

e.g. fitness for curative intent; which regimens compete and why one is preferred.

CNS risk

e.g. which risk tool and anatomical sites matter; what should NOT automatically trigger prophylaxis.

Response assessment

e.g. why this disease is suited to a particular imaging modality/criteria, and at what timepoint.

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Layer 4 — The Answer

Now the final recommendation — building directly on the reasoning above.

Be explicit: state values/results directly (e.g. "FISH negative for MYC/BCL2/BCL6") rather than implying them.

1. Diagnosis

2. Stage

3. Biology / molecular classification

4. Prognosis

5. First-line treatment

6. CNS risk

7. Response assessment

8. Missing information

Clinically relevant data not given that would change the answer if known.

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Layer 5 — The Grading

Finally — how do we tell a right answer from a dangerous one?

Critical / dangerous errors

Specific wrong answers that should fail the case outright, not just lose points.

1. Error description

Severity
minor/important/dangerous

2. Error description

Severity
minor/important/dangerous

3. Error description

Severity
minor/important/dangerous

4. Error description

Severity
minor/important/dangerous

Acceptable alternative answers

Other answers that should still receive full or partial credit.

Alternatives

Reviewer sign-off

Oncologist name

Credentials / title

Date
YYYY-MM-DD

Signature
Type name or sign if printing

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